Right Triangles & Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Proving triangles congruent**

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 10 cm. Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

2 **[Foundational]** • **Target: Proving triangles congruent**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 1$ into $y = 4x + 8$ gives $y = 4(1) + 8 = 12$.

3 **[Foundational]** • **Target: Proving triangles congruent**

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 97°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

4 **[Foundational]** • **Target: Proving triangles congruent**

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 15 cm. Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

5 **[Foundational]** • **Target: Proving triangles congruent**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 16. Substituting $x = 2$ into $y = 4x + 8$ gives $y = 4(2) + 8 = 16$.

6 **[Foundational]** • **Target: Proving triangles congruent**

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 90°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 59^\circ) = 90^\circ$.

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7 [Foundational] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving triangles congruent

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 20. Substituting $x = 3$ into $y = 4x + 8$ gives $y = 4(3) + 8 = 20$.

9 [Intermediate] • Target: Proving triangles congruent

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 83°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 66^\circ) = 83^\circ$.

10 [Intermediate] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 25 cm. Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

11 [Intermediate] • Target: Proving triangles congruent

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 24. Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

12 [Intermediate] • Target: Corresponding parts of congruent triangles

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 103°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

13 [Intermediate] • Target: Corresponding parts of congruent triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 12 cm. Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

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14 [Intermediate] • Target: Corresponding parts of congruent triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 1$ into $y = 5x + 9$ gives $y = 5(1) + 9 = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

16 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 18 cm. Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

17 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 19. Substituting $x = 2$ into $y = 5x + 9$ gives $y = 5(2) + 9 = 19$.

18 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 2.01 Lines & Angles

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 89°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 59^\circ) = 89^\circ$.

19 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

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20 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 24. Substituting $x = 3$ into $y = 5x + 9$ gives $y = 5(3) + 9 = 24$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 126°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 45^\circ) = 81^\circ$.

22 [Foundational] • Target: Angles of triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm ✓

c. 32 cm

d. 72 cm

Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

23 [Foundational] • Target: Angles of triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 10

b. 14

c. 12 ✓

d. 14

Substituting $x = 3$ into $y = 2x + 6$ gives $y = 2(3) + 6 = 12$.

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24 [Foundational] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 60° ✓

b. 70°

c. 52°

d. 126°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 66^\circ) = 60^\circ$.

25 [Foundational] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 105° ✓

b. 115°

c. 97°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$.

26 [Foundational] • Target: Review: Identifying congruent triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

27 [Foundational] • Target: Review: Identifying congruent triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 13

b. 18

c. 16 ✓

d. 19

Substituting $x = 3$ into $y = 3x + 7$ gives $y = 3(3) + 7 = 16$.

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28 [Foundational] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 66^\circ) = 84^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Angles of triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

30 [Intermediate] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 74° ✓

b. 84°

c. 66°

d. 126°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 52^\circ) = 74^\circ$.

31 [Intermediate] • Target: Angles of triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

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32 [Intermediate] • Target: Angles of triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 12

b. 16

c. 14 ✓

d. 16

Substituting $x = 4$ into $y = 2x + 6$ gives $y = 2(4) + 6 = 14$.

33 [Intermediate] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 52^\circ) = 98^\circ$.

34 [Intermediate] • Target: Review: Identifying congruent triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

35 [Intermediate] • Target: Review: Identifying congruent triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 16

b. 21

c. 19 ✓

d. 22

Substituting $x = 4$ into $y = 3x + 7$ gives $y = 3(4) + 7 = 19$.

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36 [Intermediate] • Target: Proving triangles congruent

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97° ✓

b. 107°

c. 89°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

37 [Intermediate] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving triangles congruent • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24 ✓

d. 28

Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

39 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 3.01 Triangles & Congruence

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96° ✓

b. 106°

c. 88°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

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40 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

41 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29 ✓

d. 34

Substituting $x = 4$ into $y = 5x + 9$ gives $y = 5(4) + 9 = 29$.

42 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 7.01 Circles

In ΔABC studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

43 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

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44 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 9

b. 13

c. 11 ✓

d. 13

Substituting $x = 4$ into $y = 2x + 3$ gives $y = 2(4) + 3 = 11$.

45 [Challenging] • Target: Coordinate methods and triangle congruence • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 7

b. 12

c. 10 ✓

d. 13

Substituting $x = 2$ into $y = 3x + 4$ gives $y = 3(2) + 4 = 10$.

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